

Competency

- To prove the properties of circle theoretically and experimentally and solve the related problems

Learning Outcomes

- To introduce the circle, prove the theorems on circle theoretically and verify them by induction method.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define circle - To identify the parts of the circle - To tell the relation between the line segment joining the centre of the circle and the mid-point of the chord. - To write the relationship between the chords which are equidistance form the centre of the circle
2.	Understanding (U)	- To find the length of the radius / diameter of the circle - To calculate the length of the chord
3.	Application (A)	- To verify the properties of the circle - To prove the theorems on the circle - To find the distance between the parallel lines - To show the various relations of parts of circle based on the properties and theorems
4.	High Ability (HA)	- To represent the properties of triangles, quadrilaterals and circle diagrammatically and logically. - To establish the required relations using the properties of triangles, quadrilaterals and circle in the logical way.

Required Teaching Materials/ Resources

Geometric instruments, geo-board, rubber bands, card-boards, scissors, pencils, marker, ICT tool etc.

Pre-knowledge: Circle and its parts

Teaching Activities

1. Give/ask real life examples of circular objects
2. Display circle and its parts in chart paper or card board or geo-board or ICT tools like geo-gebra and discuss
3. Experimentally verify and then prove (if possible) the following relations under discussion

- (i) The radius of the circle are equal
- (ii) The perpendicular drawn from the centre of a circle to a chord bisects the chord.
- (iii) The line perpendicular to the chord passes through the centre of the circle
- (iv) The line segment joining the mid-point of a chord and the centre of circle is perpendicular to the chord.
- (v) The line Equal chords are equidistance from the centre.
- (vi) The chords which are equidistance form the centre of a circle are equal

Solution of selected problems from Vedanta Excel in Mathematics

- 1. In the adjoining figure, O is the centre of both circle. If $PX = 3$ cm and $AQ = 8$ cm, find the length of XY .**

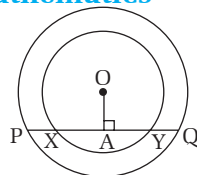
Solution:

Here, O is the centre of circle, $PX = 3$ cm and $AQ = 8$ cm

Now, (i) $AP = AQ = 8$ cm $[\because OA \perp PQ]$

(ii) $AX = AP - PX = 8$ cm $- 3$ cm $= 5$ cm

(iii) $XY = 2 AX = 2 \times 5$ cm $= 10$ cm $[\because OA \perp XY]$



- 2. In the figure alongside, O is the centre of a circle. $AB = 20$ cm, $CD = 16$ cm and $AB \parallel CD$. Find the distance between AB and CD .**

Solution:

Here, O is the centre of circle, $AB = 20$ cm, $CD = 16$ cm and $AB \parallel CD$

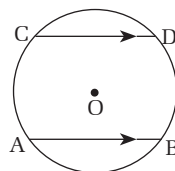
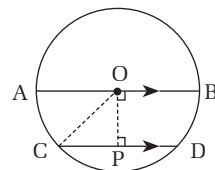
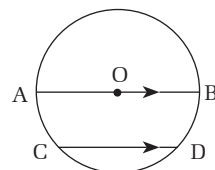
Construction: $OP \perp CD$ is drawn and OC is joined

Then, $OC =$ radius $= 10$ cm

Now, $CP = \frac{1}{2} CD = 8$ cm $[\because OP \perp CD]$

In right angled triangle COP ; $OP = \sqrt{OC^2 - CP^2} = \sqrt{10^2 - 8^2} = 6$ cm

Thus, the distance between the AB and CD is 6 cm



- 3. In the given figure, O is the centre of a circle. AB and CD are two parallel chords of lengths 16 cm and 12 cm respectively. If the radius of the circle is 10 cm, find the distance between the chords.**

Solution:

Here, O is the centre of circle, $AB = 16$ cm, $CD = 12$ cm, radius $= 10$ cm and $AB \parallel CD$

Construction: $OM \perp AB$ and $ON \perp CD$ are drawn and, A and C are joined to O.

Now, $AM = \frac{1}{2} AB = 8$ cm $[\because OM \perp AB]$

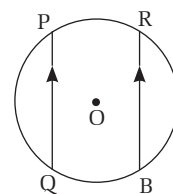
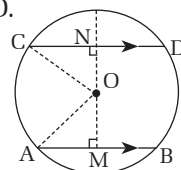
In right angled triangle AOM ; $OM = \sqrt{OA^2 - AM^2} = \sqrt{5^2 - 4^2} = 3$ cm

Again, $CN = \frac{1}{2} CD = 6$ cm $[\because ON \perp CD]$

In right angled triangle CON ; $ON = \sqrt{OC^2 - CN^2} = \sqrt{5^2 - 3^2} = 4$ cm

Now, $MN = OM + ON = 6$ cm $+ 8$ cm $= 14$ cm

Thus, the distance between the AB and CD is 14 cm.



- 4. In the adjoining figure, O is the centre of a circle, PQ and RS are two equal and parallel chords. If the radius of the circle is 5 cm and the distance between the chords is 8 cm, find the length of the chords.**

Solution:

Here, O is the centre of circle, $PQ = RS$ and $PQ \parallel RS$

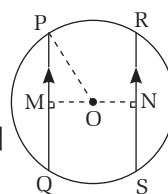
Construction: $OM \perp PQ$ and $ON \perp RS$ are drawn and, P and O joined.

Now, $OM = \frac{1}{2} MN = 4$ cm [Equal chords are equidistance from the centre]

Again, in right angled triangle POM; $\sqrt{OP^2 - OM^2} = \sqrt{5^2 - 4^2} = 3$ cm

$\therefore PQ = 2PM = 2 \times 3$ cm = 6 cm

Hence, the length of each of chords PQ and RS is 6 cm.



5. In a circle of radius 5 cm, AB and CD are two parallel chords of length 8 cm and 6 cm respectively. Calculate the distance between the chords, if they are

(i) on the same side of the centre (ii) on the opposite side of the centre.

Solution:

Here, O is the centre of circle, radius = 5 cm, $AB = 8$ cm, $CD = 6$ cm and $AB \parallel CD$

(i) If AB and CD lie on the same side of the centre

Construction: $OM \perp AB$ and $ON \perp CD$ are drawn and, A and C are joined to O.

Now, $AM = \frac{1}{2} AB = 4$ cm [$\because OM \perp AB$]

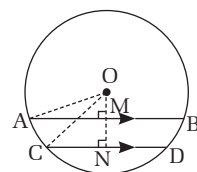
In right angled triangle AOM; $OM = \sqrt{OA^2 - AM^2} = \sqrt{5^2 - 4^2} = 3$ cm

Again, $CN = \frac{1}{2} CD = 3$ cm [$\because ON \perp CD$]

In right angled triangle CON; $ON = \sqrt{OC^2 - CN^2} = \sqrt{5^2 - 3^2} = 4$ cm

Now, $MN = ON - OM = 4$ cm - 3 cm = 1 cm

Thus, the distance between the AB and CD is 1 cm.



(ii) If AB and CD lie on the opposite side of the centre

Construction: $OM \perp AB$ is drawn and MO is produced to meet CD at N. So, $ON \perp CD$ [$\because AB \parallel CD$]

Now, $AM = \frac{1}{2} AB = 4$ cm [$\because OM \perp AB$]

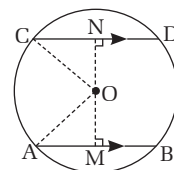
In right angled triangle AOM; $OM = \sqrt{OA^2 - AM^2} = \sqrt{5^2 - 4^2} = 3$ cm

Again, $CN = \frac{1}{2} CD = 3$ cm [$ON \perp CD$]

In right angled triangle CON; $ON = \sqrt{OC^2 - CN^2} = \sqrt{5^2 - 3^2} = 4$ cm

Now, $MN = OM + ON = 3$ cm + 4 cm = 7 cm

Thus, the distance between the AB and CD is 7 cm.



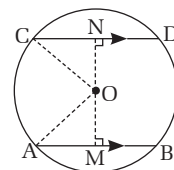
6. AB and CD are two parallel chords of a circle such that $AB = 10$ cm and $CD = 24$ cm. If the chords are on the opposite sides of the centre and the distance between them is 17 cm, find the radius of the circle.

Solution:

Here, O is the centre of circle, $AB = 10$ cm, $CD = 24$ cm and $AB \parallel CD$

AB and CD lie on the opposite side of the centre

Construction: $OM \perp AB$ is drawn and MO is produced to meet CD at N. So,



$$ON \perp CD \quad [\because AB \parallel CD]$$

Let the radius of the circle $OA = OC = r$ and $OM = x$ then

$$ON = MN - OM = (17 - x) \text{ cm}$$

$$\text{Now, } AM = \frac{1}{2} AB = 5 \text{ cm} \quad [\because OM \perp AB]$$

$$CN = \frac{1}{2} CD = 12 \text{ cm} \quad [\because ON \perp CD]$$

$$\text{In right angled triangle } AOM; OA^2 = OM^2 + AM^2 = x^2 + 5^2 = x^2 + 25$$

$$\text{In right angled triangle } CON; OC^2 = ON^2 + CN^2 = (17 - x)^2 + 12^2 = 433 - 34x + x^2$$

$$\text{As } OA = OC, OA^2 = OC^2 \quad \text{or, } x^2 + 25 = 433 - 34x + x^2 \quad \therefore x = 12$$

$$\text{Again, } OA^2 = r^2 = x^2 + 25 = 12^2 + 25 = 169 \quad \therefore r = 13$$

Hence, the radius of the circle is 13 cm.

7. *In the given figure, O is the centre of the circle, chords MN and RS are intersecting at P. If OP is the bisector of $\angle MPR$, prove that $MN = RS$.*

Solution:

Given: O is the centre of the circle, chords MN and RS intersect at P and

OP is the bisector of $\angle MPR$

To prove: $MN = RS$

Construction: $OA \perp MN$ and $OB \perp RS$ are drawn.

Proof:

(1) In $\triangle AOP$ and $\triangle BOP$

(i) $\angle OAP = \angle OBP$ (A) [Both are right angles]

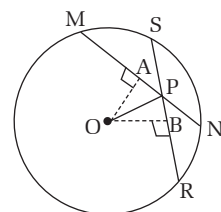
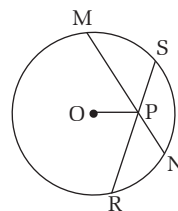
(ii) $\angle OPA = \angle OPB$ (A) [OP is the bisector of $\angle MPR$]

(iii) $OP = OP$ (S) [Common side]

(2) $\triangle AOP \cong \triangle BOP$ [By A.A.S axiom]

(3) $OA = OB$ [Corresponding sides of congruent triangles]

(4) $MN = RS$ [From (3), chords equidistant from centre]



8. *In the figure alongside, PQ and RS are two chords intersecting at T in a circle with centre O. If OT is the bisector of $\angle PTR$, prove that (i) $PT = RT$ (ii) $ST = TQ$*

Solution:

Given: O is the centre of the circle, chords PQ and RS intersect at T and

OT is the bisector of $\angle PTR$

To prove: (i) $PT = RT$ (ii) $ST = TQ$

Construction: $OA \perp PQ$ and $OB \perp RS$ are drawn.

Proof:

(1) In $\triangle AOT$ and $\triangle BOT$

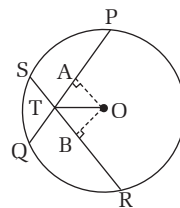
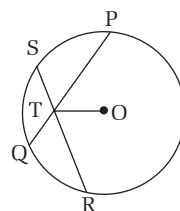
(i) $\angle OAT = \angle OBT$ (A) [Both are right angles]

(ii) $\angle OTA = \angle OTB$ (A) [OT is the bisector of $\angle PTR$]

(iii) $OT = OT$ (S) [Common side]

(2) $\triangle AOT \cong \triangle BOT$ [By A.A.S axiom]

(3) $OA = OB$ and $AT = BT$ [Corresponding sides of congruent triangles]



- (4) $PQ = RS$ [From (3), chords equidistance from centre]
 (5) $AP = BR$ [From (4) and $OA \perp PQ$, $OB \perp RS$]
 (6) $PT = RT$ [Adding (3) and (5), $AT + AP = BT + BR$]
 (7) $ST = TQ$ [Subtracting (6) from (4), $RS - RT = PQ - PT$]

9. In the given figure, O is the centre of the circle. Two equal chords AB and CD intersect each other at E. Prove that (i) $AE = CE$ (ii) $BE = DE$

Solution:

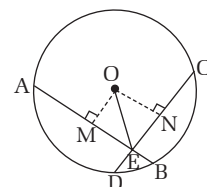
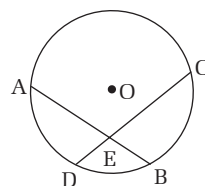
Given: O is the centre of the circle, equal chords AB and CD intersect at E
 at E

To prove: (i) $AE = CE$ (ii) $BE = DE$

Construction: $OM \perp AB$ and $ON \perp CD$ are drawn, O and E are joined

Proof:

- (1) In $\triangle MOE$ and $\triangle NOE$
 (i) $\angle OME = \angle ONE$ (R) [Both are right angles]
 (ii) $OE = OE$ (H) [Common side]
 (iii) $OM = ON$ (S) [Equal chords are equidistance from the centre]
 (2) $\triangle MOE \cong \triangle NOE$ [By R.H.S axiom]
 (3) $ME = NE$ [Corresponding sides of congruent triangles]
 (4) $AM = CN$ [AB = CD and $OM \perp AB$, $ON \perp CD$]
 (5) $AE = CE$ [Adding (3) and (4), $ME + AM = NE + CN$]
 (6) $AB = CD$ [Given]
 (7) $BE = DE$ [Subtracting (5) from (6), $AB - AE = CD - CE$]



10. In the figure, L and M are the mid-points of two equal chords AB and CD of a circle with centre O. Prove that

(i) $\angle OLM = \angle OML$ (ii) $\angle ALM = \angle CML$

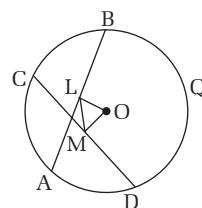
Solution:

Given: O is the centre of the circle, L and M are the mid-points of equal chords AB and CD respectively

To prove: (i) $\angle OLM = \angle OML$ (ii) $\angle ALM = \angle CML$

Proof:

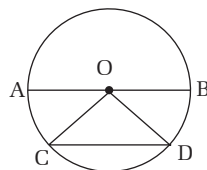
- (1) $OL \perp AB$, $OM \perp CD$ [AL = BL and AM = DM]
 (2) $\angle OLA = \angle OMC$ [From (1), both are right angles]
 (3) $OL = OM$ [Equal chords are equidistance from the centre]
 (4) $\angle OLM = \angle OML$ [From (3)]
 (5) $\angle ALM = \angle CML$ [Subtracting (4) from (2)]



11. In the adjoining figure, AB is the diameter of a circle with centre O. If chord $CD \parallel AB$, prove that $\angle AOC = \angle BOD$.

Solution:

Given: O is the centre of the circle, AB is the diameter and chord $CD \parallel AB$

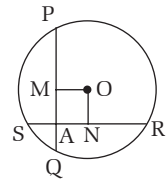


To prove: $\angle AOC = \angle BOD$

Proof:

- | | |
|-------------------------------|-------------------------------|
| (1) $\angle OCD = \angle ODC$ | [OC = OD] |
| (2) $\angle OCD = \angle AOC$ | [AB//CD and alternate angles] |
| (3) $\angle ODC = \angle BOD$ | [AB//CD and alternate angles] |
| (4) $\angle AOC = \angle BOD$ | [From (1), (2) and (3)] |

12. In the given figure, equal chords PQ and RS of a circle with centre O intersect each other at right angle at A. If M and N are the mid-points of PQ and RS respectively, prove that OMAN is a square.



Solution:

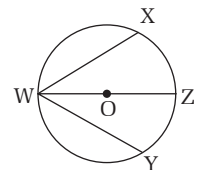
Given: O is the centre of the circle, equal chords PQ and RS intersect at right angle at A. M and N are the midpoints of PQ and RS

To prove: OMAN is a square

Proof:

- (1) $OM \perp PQ$ [OM joins the centre O and mid-point M of the chord PQ]
- (2) $ON \perp RS$ [ON joins the centre O and mid-point N of the chord RS]
- (3) $\angle MAN = 90^\circ$ [Given]
- (4) $\angle MAN = 90^\circ$ [Remaining angle of the quadrilateral OMAN]
- (5) $OM = ON$ [Equal chords of a circle are equidistant from the centre]
- (6) OMAN is a square [From (1), (2), (3), (4) and (5)]

13. In the adjoining figure, two chords WX and WY are equally inclined to the diameter at their point of intersection. Prove that the chords are equal.



Solution:

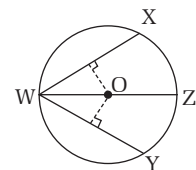
Given: O is the centre of the circle, chords WX and WY are equally inclined to the diameter WZ at W.

To prove: Chords WX and WY are equal

Construction: $OM \perp WX$ and $ON \perp WY$ are drawn

Proof:

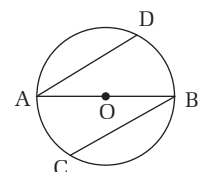
- (1) In $\triangle MOW$ and $\triangle NOW$
 - (i) $\angle OMW = \angle NOW$ (A) [Both are right angles]
 - (ii) $\angle OWM = \angle ONW$ (A) [Given]
 - (iii) $OW = OW$ (S) [Common side]
- (2) $\triangle MOW \cong \triangle NOW$ [By A.A.S axiom]
- (3) $OM = ON$ [Corresponding sides of congruent triangles]
- (4) $WX = WY$ [From (3) and $OM \perp WX$, $ON \perp WY$]



14. In the figure alongside, AB is a diameter. Two chords AD and BC are equal. Prove that $AD \parallel BC$.

Solution:

Given: O is the centre of the circle, AB is the diameter, and chords AD and BC are equal



To prove: $AD \parallel BC$

Construction: $OM \perp AD$ and $ON \perp BC$ are drawn

Proof:

(1) In $\triangle OMA$ and $\triangle ONB$

(i) $\angle OMA = \angle ONB$ (R) [Both are right angles]

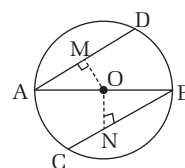
(ii) $OA = OB$ (H) [Radii]

(iii) $OM = ON$ (S) [Equal chords are equidistance from the centre]

(2) $\triangle OMA \cong \triangle ONB$ [By R.H.S axiom]

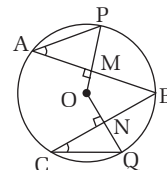
(3) $\angle OAM = \angle OBN$ [Corresponding angles of congruent triangles]

(4) $AD \parallel BC$ [From (3), alternate angles are equal]



15. In the given figure, AB and BC are equal chords of the circle with centre O . If $OM \perp AB$, $ON \perp BC$, OM and ON are produced to meet the circumference at P and Q respectively. Prove that:

(i) $AP = CQ$ (ii) $\angle PAM = \angle QCN$.



Solution:

Given: O is the centre of the circle, chords AB and BC are equal, $OM \perp AB$, $ON \perp BC$, OM and ON are produced to meet the circumference at P and Q respectively.

To prove: (i) $AP = CQ$ (ii) $\angle PAM = \angle QCN$

Proof:

(1) $OP = OQ$ [Radii]

(2) $OM = ON$ [Equal chords are equidistance from the centre]

(3) $PM = QN$ [Subtracting (2) from (1)]

(4) In $\triangle AMP$ and $\triangle CNQ$

(i) $PM = QN$ (S) [From (3)]

(ii) $\angle AMP = \angle CNQ$ (A) [Given]

(iii) $AM = CN$ (S) [$AB = BC$, $OM \perp AB$ and $ON \perp BC$]

(5) $\triangle AMP \cong \triangle CNQ$ [By S.A.S axiom]

(6) $AP = CQ$ and $\angle PAM = \angle QCN$ [Corresponding parts of congruent triangles]

16. Two equal chords AB and CD of a circle with centre O are produced to meet at E , as shown in the given figure. Prove that $BE = DE$ and $AE = CE$.

Solution:

Given: O is the centre of the circle, equal chords AB and CD are produced to meet at E

To prove: (i) $BE = DE$ (ii) $AE = CE$

Construction: $OM \perp AB$ and $ON \perp CD$ are drawn

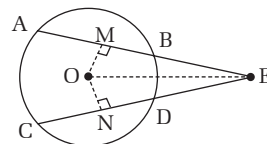
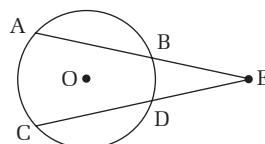
Proof:

(1) In $\triangle MOE$ and $\triangle NOE$

(i) $\angle OME = \angle ONE$ (R) [Both are right angles]

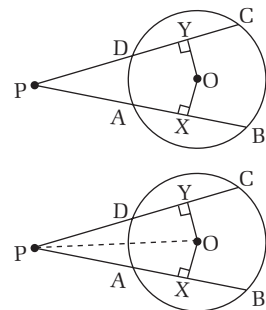
(ii) $OE = OE$ (H) [Common side]

(iii) $OM = ON$ (S) [Equal chords are equidistance from the centre]



- (2) $\triangle MOE \cong \triangle NOE$ [By R.H.S axiom]
 (3) $ME = NE$ [Corresponding sides of congruent triangles]
 (4) $MB = ND$ [AB=CD, $OM \perp AB$ and $ON \perp CD$]
 (5) $BE = DE$ [Subtracting (4) from (3), $ME - MB = NE - ND$]
 (6) $AM = CN$ [AB=CD, $OM \perp AB$ and $ON \perp CD$]
 (7) $AE = CE$ [Adding (3) and (6), $ME + AM = NE + CN$]

17. In the given figure, O is the centre of circle ABCD.
If $OY \perp PC$, $OX \perp PB$ and $OX = OY$, prove that $PB = PC$.



Solution:

Given: O is the centre of the circle, $OY \perp PC$, $OX \perp PB$ and $OX = OY$

To prove: $PB = PC$

Construction: O and P are joined

Proof:

- (1) In $\triangle POX$ and $\triangle POY$
 (i) $\angle OXP = \angle OYP$ (R) [Both are right angles]
 (ii) $OP = OP$ (H) [Common side]
 (iii) $OX = OY$ (S) [Given]
 (2) $\triangle POX \cong \triangle POY$ [By R.H.S axiom]
 (3) $PX = PY$ [Corresponding sides of congruent triangles]
 (4) $AB = CD$ [$OX \perp AB$, $OY \perp CD$ and $OX = OY$]
 (5) $BX = CX$ [From (4), $AB = CD$ and $OX \perp AB$, $OY \perp CD$]
 (6) $PB = PC$ [Adding (3) and (5), $PX + PB = PY + PC$]

18. In the given figure, ABC is a triangle in which $AB = AC$. Also a circle passing through B and C intersects the sides AB and AC at the points D and E respectively. Prove that $AD = AE$.

Solution:

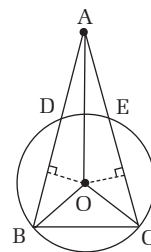
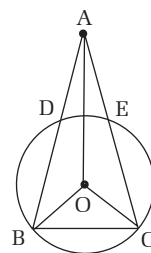
Given: In triangle ABC, $AB = AC$. The circle passing through B and C intersects the side AB and AC at D and E respectively.

To prove: $AD = AE$

Construction: $OP \perp AD$ and $OQ \perp CE$ are drawn

Proof:

- (1) $AB = AC$ [Given]
 (2) $\angle ABC = \angle ACB$ [From (1)]
 (3) $\angle OBC = \angle OCB$ [$OB = OC$, the radii]
 (4) $\angle ABO = \angle ACO$ [Subtracting (3) from (2)]
 (5) In $\triangle POB$ and $\triangle QOC$
 (i) $\angle OPB = \angle OQC$ (A) [Both are right angles]
 (ii) $\angle PBO = \angle QCO$ (A) [From (4), $\angle ABO = \angle ACO$]
 (iii) $OB = OC$ (S) [Radii]
 (6) $\triangle POB \cong \triangle QOC$ [By A.A.S axiom]



- (7) $OP = OQ$ [Corresponding sides of congruent triangles]
 (8) $BD = CE$ [$OP \perp BD$, $OQ \perp CE$ and $OP = OQ$]
 (9) $AD = AE$ [Subtracting (8) from (1), $AB - BD = AC - CE$]

19. In the given figure, O is the centre of a circle, AB the diameter, AC the chord and $OM \perp AC$. Prove that: (i) $OM \parallel BC$ (ii) $BC = 2 OM$

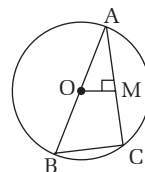
Solution:

Given: O is the centre of circle, AB the diameter, AC the chord and $OM \perp AC$

To prove: (i) $OM \parallel BC$ (ii) $BC = 2 OM$

Proof:

- (1) $AM = MC$ [$OM \perp AC$ and OM bisects the chord AC]
 (2) $OA = OB$ [Radii]
 (3) $OM \parallel BC$ and $BC = 2 OM$ [OM joins the mid-points of AB and AC]



20. In the adjoining figure, P is the centre of a circle. If $PQ \parallel BC$, prove that: (i) $AC = 2AQ$ (ii) $PQ \perp AC$.

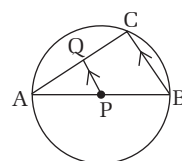
Solution:

Given: P is the centre of circle, AB the diameter, C is a point on the circumference and $PQ \parallel BC$

To prove: (i) $AC = 2AQ$ (ii) $PQ \perp AC$

Proof:

- (1) $AP = BP$ [Radii]
 (2) $AQ = QC$ i.e., $AC = 2AQ$ [In $\triangle ABC$; $AP = BP$ and $PQ \parallel BC$]
 (3) $PQ \perp AC$ [PQ joins the mid-point of chord AC and the centre of the circle]



21. In the given figure, two circles with centres P and Q intersect at A and B . Prove that the line joining the two centres the circles is the perpendicular bisector of the common chord.

Solution:

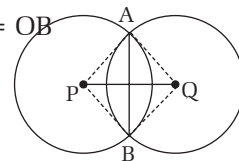
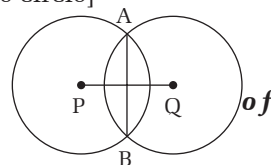
Given: Circles with centres P and Q intersect at A and B

To prove: PQ is the perpendicular bisector of AB i.e., $PO \perp AB$ and $OA = OB$

Construction: A and B are joined to P and Q

Proof:

- (1) In $\triangle PAQ$ and $\triangle PBQ$
 (i) $AP = BP$ (S) [Radii of circle with centre P]
 (ii) $AQ = BQ$ (S) [Radii of circle with centre Q]
 (iii) $PQ = PQ$ (S) [Common side]
 (2) $\triangle PAQ \cong \triangle PBQ$ [By S.S.S axiom]
 (3) $\angle APQ = \angle BPQ$ [Corresponding angles of congruent triangles]
 (4) $PO \perp AB$ and $OA = OB$ [In $\triangle PAB$, $PA = PB$ and $\angle APQ = \angle BPQ$]



22. In the given figure, P and Q be the centres of two intersecting circles and $AB \parallel PQ$. Prove that $AB = 2 PQ$.

Solution:

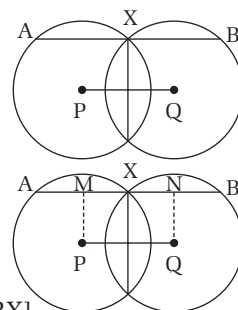
Given: Circles with centres P and Q intersect at X and Y and $AB \parallel PQ$

To prove: $AB = 2 PQ$

Construction: $PM \perp AX$ and $QN \perp BX$ are drawn

Proof:

- (1) $PQNM$ is a rectangle [$AB \parallel PQ$ and by construction]
- (2) $AX = 2MX$ and $BX = 2NX$ [$PM \perp AX$, $QN \perp BX$ and bisect the AX and BX]
- (3) $AB = AX + BX$ [Whole part axiom]
- (4) $AB = 2(MX + NX) = 2MN$ [From (2) and (3)]
- (5) $AB = 2PQ$ [From (4), $MN = PQ$; the opposite sides of rectangle]



23. In the given figure, two circles with centres P and Q are intersecting at A and B . If MN is parallel to common chord AB , prove that

(i) $MC = ND$

(ii) $MD = NC$

Solution:

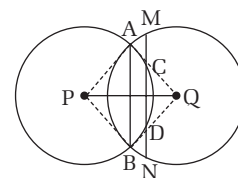
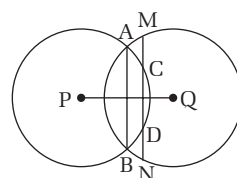
Given: Circles with centres P and Q intersect at A and B and $MN \parallel AB$

To prove: (i) $MC = ND$ (ii) $MD = NC$

Construction: A and B are joined to P and Q

Proof:

- (1) In $\triangle PAQ$ and $\triangle PBQ$
 - (i) $AP = BP$ (S) [Radii of circle with centre P]
 - (ii) $AQ = BQ$ (S) [Radii of circle with centre Q]
 - (iii) $PQ = PQ$ (S) [Common side]
- (2) $\triangle PAQ \cong \triangle PBQ$ [By S.S.S axiom]
- (3) $\angle APQ = \angle BPQ$ [Corresponding angles of congruent triangles]
- (4) $PO \perp AB$ and $OA = OB$ [In $\triangle PAB$, $PA = PB$ and $\angle APQ = \angle BPQ$]
- (5) $PQ \perp CD$ [From (4), $AB \parallel MN$]
- (6) $CX = DX$ [$PX \perp CD$ and PX bisects CD]
- (7) $MX = NX$ [$QX \perp MN$ and QX bisects MN]
- (8) $MC = ND$ [Subtracting (6) from (7), $MX - CX = NX - DX$]
- (9) $MD = NC$ [Adding CD in (8), $MC + CD = ND + CD$]



24. Prove that the diameter of a circle perpendicular to one of the two parallel chords of a circle is perpendicular to the other and bisects it.

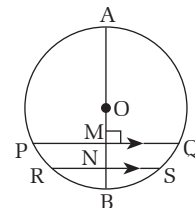
Solution:

Given: O is the centre of circle, AB the diameter, chord PQ is perpendicular to the diameter AB and $PQ \parallel RS$

To prove: AB is perpendicular bisector of RS

Proof:

- (1) $ON \perp RS$ [$OM \perp PQ$ and $PQ \parallel RS$]
- (2) $RN = NS$ [$ON \perp RS$ and ON bisects RS]
- (3) AB is perpendicular bisector of RS [From (1) and (2)]



25. Prove that the line joining the mid-points of two parallel chords of a circle passes through the centre of the circle.

Solution:

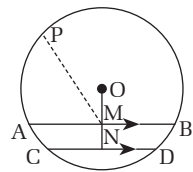
Given: O is the centre of circle, $AB \parallel CD$, M and N are the mid-points of the chords AB and CD respectively

To prove: MN passes through the centre O

Construction: P is a point such that $PM \perp AB$ and O does not lie on PM

Proof:

- (1) $OM \perp AB$ [OM joins the mid-point of AB]
- (2) $PM \perp AB$ [By assumption]
- (3) $\angle OMB = \angle PMB$ [From (1) and (2)]
- (4) PM passes through O [From (3)]
- (5) $ON \perp CD$ and $PN \perp CD$ [ON joins the mid-point of CD and $AB \parallel CD$]
- (6) MN passes through the centre O [From (4) and (5)]



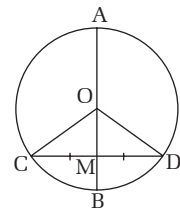
26. Prove that a diameter of a circle which bisects a chord of the circle also bisects the angle subtended by the chord at the centre of the circle.

Solution:

Given: O is the centre of circle, AB is the diameter, $CM = DM$

To prove: $\angle COM = \angle DOM$

- (1) In $\triangle COM$ and $\triangle DOM$
 - (i) $OC = OD$ (S) [Radii of the circle]
 - (ii) $CM = DM$ (S) [Given]
 - (iii) $OM = OM$ (S) [Common side]
- (2) $\triangle COM \cong \triangle DOM$ [By S.S.S axiom]
- (3) $\angle COM = \angle DOM$ [Corresponding angles of congruent triangles]



27. Of two chords, the chord nearer to the centre of circle is longer.

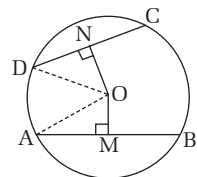
Solution:

Given: O is the centre of circle; AB and CD are two chords such that $AB > CD$. $OM \perp AB$ and $ON \perp CD$

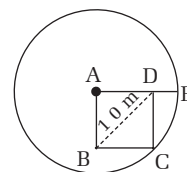
To prove: $OM < ON$

Construction: A and C are joined to O

- (1) $AM = \frac{1}{2} AB$ and $DN = \frac{1}{2} CD$ [OM $\perp$ AB and OM bisects AB, ON $\perp$ CD and ON bisects CD]
- (2) $OA^2 = OM^2 + AM^2$ and $OD^2 = ON^2 + DN^2$ [Using Pythagoras theorem in rt. $\triangle AOM$ and $\triangle DON$]
- (3) $OM^2 + AM^2 = ON^2 + DN^2$ [From (2)]
i.e., $OM^2 - ON^2 = DN^2 - AM^2$
- (4) $DN < AM$ or, $DN^2 < AM^2$ [$AB > CD$ i.e., $CD < AB$ and from (1)]
- (5) $OM^2 < ON^2$ or, $OM < ON$ [From (3) and (4)]



28. Five students Amrit, Bibika, Chandani, Dipesh and Elina are playing in a circular meadow. Amrit is at the centre, Bibika and Dipesh are inside the boundary line. Similarly, Chandani and Elina are on the boundary of the meadow. If Amrit, Bibika, Chandani and Dipesh form a rectangle and the distance between Bibika and Dipesh is 10 m, find the distance between Amrit and Elina.

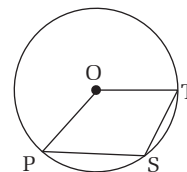


Solution:

- (i) $AC = BD = 10$ m [Diagonals of rectangle are equal]
 (ii) $AC = AE = 10$ m [Radii of the circle]

Hence, the distance between Amrit and Elina is 10 m.

29. Three students Pooja, Shaswat and Triptee are playing a game by standing on the circumference of a circle of radius 25 feet drawn in a park. Pooja throws a ball to Shaswat and Shaswat to Triptee and Triptee to Pooja. What is the distance between Pooja and Triptee when the distance between Pooja and Shaswat and the distance between Shaswat and Triptee is 30 feet each?



Solution:

Here, $OP = OT = OS = 25$ feet, $PS = ST = 30$ feet

Since, $OP = OT$ and $PS = ST$. So, $OPST$ is a kite in which diagonal OS bisects the diagonal PT at M at a right angle.

Let $OM = x$ feet then $MS = OS - OM = (25 - x)$ feet

Now,

From rt. $\triangle OPM$, $PM^2 = OP^2 - OM^2 = 25^2 - x^2 = 625 - x^2$... (i)

From rt. $\triangle PMS$, $PM^2 = PS^2 - MS^2 = 30^2 - (25 - x)^2 = 900 - (625 - 50x + x^2)$
 $= 275 + 50x - x^2$... (ii)

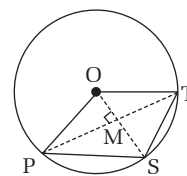
From (i) and (ii), we get

$$625 - x^2 = 275 + 50x - x^2 \quad \therefore x = 7$$

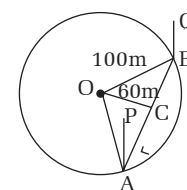
Also, from (i); $PM^2 = 625 - 7^2 = 576 \quad \therefore PM = 24$ feet

Again, $PT = 2 \times PM = 2 \times 24$ feet = 48 feet

Hence, the distance between Pooja and Triptee is 48 feet.



30. The diameter of a circular ground with centre at O is 200 m. Two vertical poles P and Q are fixed at the two points in the circumference of the ground. Find the length of a rope required to tie the poles tightly at a distance of 60 m from the centre of the ground.



Solution:

Let the poles P and Q are fixed at the points A and B on the circumference and $OC \perp AB$.

Then the required length of rope to tie the poles is AB . and diameter = 200 m

$\therefore$ radius = $OA = OB = 100$ m and $OC = 60$ m

Now,

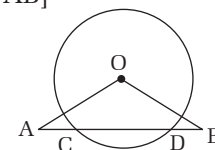
From rt. $\triangle OBC$, $BC^2 = OB^2 - OC^2 = 100^2 - 60^2 = 6400 \quad \therefore BC = 80$ m

Also, $AB = 2BC = 2 \times 80$ m = 160 m [$OC \perp AB$ and OC bisects AB]

31. In the adjoining figure, OAB is an isosceles triangle and a circle with O as the centre cuts AB at C and D . Prove that $AC = DB$.

Solution:

Given: O is the centre of circle; OAB is an isosceles triangle

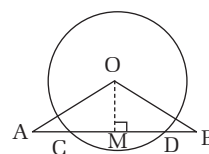


To prove: $AC = BD$

Construction: $OM \perp AB$ is drawn

Proof:

- | | |
|--|--|
| (1) In $\triangle AOM$ and $\triangle BOM$ | |
| (i) $\angle OMA = \angle OMB$ (A) | [Both are right angles] |
| (ii) $\angle OAM = \angle OBM$ (A) | [Base angles of isosceles triangle] |
| (iii) $OA = OB$ (S) | [Given] |
| (2) $\triangle AOM \cong \triangle BOM$ | [By A.A.S axiom] |
| (3) $AM = BM$ | [Corresponding sides of congruent triangles] |
| (4) $CM = BD$ | [$OM \perp CD$ and OM bisects CD] |
| (5) $AC = BD$ | [Subtracting (4) from (3), $AM - CM = BM - BD$] |



32. In the figure alongside, MN is the diameter of a circle with centre O . If $BD = CD$, prove that $\angle OAD = \angle OCD$.

Solution:

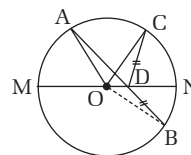
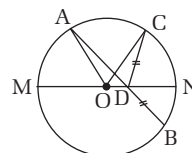
Given: O is the centre of circle; MN is the diameter, $BD = CD$

To prove: $\angle OAD = \angle OCD$

Construction: O and B are joined

Proof:

- | | |
|--|---|
| (1) In $\triangle COD$ and $\triangle BOD$ | |
| (i) $OC = OB$ (S) | [Radii of the circle] |
| (ii) $OD = OD$ (S) | [Common side] |
| (iii) $CD = BD$ (S) | [Given] |
| (2) $\triangle COD \cong \triangle BOD$ | [By S.S.S axiom] |
| (3) $\angle OCD = \angle OBD$ | [Corresponding angles of congruent triangles] |
| (4) $\angle OAD = \angle OBD$ | [$OA = OB$] |
| (5) $\angle OAD = \angle OCD$ | [From (3) and (4)] |



33. In the figure alongside, A and B are the centres of two intersecting circles. If CD intersects AB perpendicularly at P , prove that

(i) $CM = DN$

(ii) $CN = DM$

Solution:

Given: A and B are the centres of the intersecting circles. CD intersects AB perpendicularly

To prove: (i) $CM = DN$

(ii) $CN = DM$

Proof:

- | | |
|---------------|--|
| (1) $PM = PN$ | [$AP \perp MN$ and AP bisects MN] |
| (2) $PC = PD$ | [$BP \perp CD$ and BP bisects CD] |
| (3) $CM = DN$ | [Subtracting (1) from (2); $PC - PM = PD - PN$] |
| (4) $CN = DM$ | [Adding MN in (3), $CM + MN = DN + MN$] |

